SOFTWARE REQUIREMENTS SPECIFICATION

for

PeerBridge — Intercollege Peer Learning and Mentorship Platform

A web application where senior college students conduct learning sessions, quizzes, and mentorship for junior students, with points, leaderboards, and ratings to track participation.

Item	Details
Programme	[Your Programme / Department Name]
Project title	PeerBridge
Team	Group of 3 students
Document type	Software Requirements Specification (SRS)
Version	1.0 (Draft)
Date	10 October 2026
Status	Draft, for review

Document Control

Version	Date	Author	Description
0.1	10-Oct-2026	Project team	Initial draft from problem statement
1.0	10-Oct-2026	Project team	Full SRS with module requirements

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1. Introduction

1.1 Purpose

This document specifies the requirements of the PeerBridge web application. It is intended for the project team, the course mentor/guide, and anyone reviewing the project.

1.2 Problem Statement

Junior engineering students (1st and 2nd year) often lack structured guidance for difficult subjects and placement preparation. Senior students (3rd and 4th year) already have this knowledge and experience, but there is no organised platform connecting the two groups. Informal channels like WhatsApp groups are untraceable and disappear quickly. Paid platforms are generic and not aligned with a specific college's syllabus or culture.

PeerBridge is a web platform that formalises senior-to-junior peer learning within a college. Seniors create and conduct learning sessions, share resources, and set quizzes. Juniors discover sessions, attend them, attempt quizzes, and rate mentors. A points and leaderboard system makes participation and contribution measurable for both sides.

1.3 Scope

In scope	Out of scope
College students (any department/year)	School students
Six modules: Authentication, Sessions, Quiz, Leaderboard, Ratings, Resources	Live video calling inside the app
Junior and Senior roles	Automatic Google Meet/Calendar creation
Points and gamification	Payments, certificates, discussion forums
Web application (desktop and mobile browser)	Native mobile app
Mentor pastes a meeting link (Google Meet/Zoom/Teams)	Real-time chat between users

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
Junior	A 1st or 2nd year student; primarily a learner on the platform
Senior / Mentor	A 3rd or 4th year student; both a learner and a session creator
Session	A scheduled learning class created by a senior, with a meeting link
Registration	A junior signing up to attend a specific session
Quiz	A set of questions linked to a session, created by the mentor
Points	Numeric reward given to a user for an action (e.g., completing a quiz)
Leaderboard	A ranked list of users based on points (students) or rating (mentors)
Badge	A title given to a student based on their point range
Rating	A 1-to-5 score a junior gives a mentor after a session
JWT	JSON Web Token — used to keep a user logged in securely

Term	Meaning
API	Application Programming Interface — how the frontend and backend talk to each other
FR / NFR	Functional Requirement / Non-Functional Requirement

1.5 References

Full links and sources are listed in Appendix A. Key sources:

- Original project idea document (team-authored)
- MERN stack documentation (React, Express, MongoDB, Node.js)

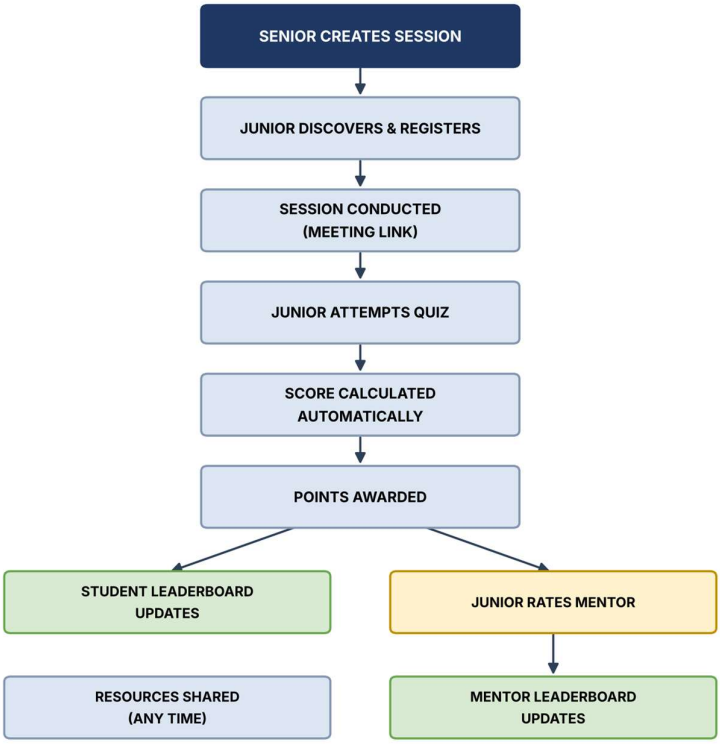
1.6 Document Overview

Section 2 gives the overall description of the system. Section 3 lists detailed requirements for each of the six modules. Section 4 covers requirements that apply to the whole system. Section 5 covers the technical stack and data design. Appendix A lists open issues the team still needs to decide.

2. Overall Description

2.1 Product Perspective

PeerBridge is a new, standalone web application with six connected modules. A student's actions in one module affect what they see in another — for example, completing a quiz adds points that show up on the Leaderboard, and attending a session unlocks the ability to rate that session's mentor.



Resources can be shared and browsed by any logged-in user at any point in this flow.

Figure 1: System flow across modules

2.2 Product Functions

Module	Main function	Output
1. Authentication & Profile	Register, log in, manage role-based access and profile	Logged-in user with a known role (junior/senior)
2. Session Management	Senior creates sessions; junior discovers and registers	A scheduled session with registered students
3. Quiz	Senior creates a quiz for a session; junior attempts it	Auto-calculated score and points
4. Leaderboard & Gamification	Tracks points, ranks students and mentors	Student leaderboard, mentor leaderboard, badges
5. Ratings & Feedback	Junior rates the mentor after a session	Mentor's average rating and feedback history
6. Resource Sharing	Users share links to notes, videos, and material	Searchable library of study resources

2.3 User Classes

User	Description	Main actions
Junior Student	1st or 2nd year student	Register, log in, browse sessions, register for sessions, attempt quizzes, rate mentors, view leaderboard, browse resources
Senior Student / Mentor	3rd or 4th year student	All junior actions, plus: create/edit/delete sessions, create quizzes, view registered students, view own ratings

An Admin role is listed as future scope — see Appendix A, OI-1.

2.4 Stakeholders

Stakeholder	Interest
Junior students	Easier access to guidance from experienced peers
Senior students	A way to mentor, build their profile, and be recognised for teaching
College / Department	A structured, low-cost peer-learning system
Course mentor / guide	Project quality, completeness, and evaluation
Project team (3 students)	Design, development, and testing

2.5 Operating Environment

- Web browser: latest Chrome, Edge, or Firefox
- Devices: laptop, tablet, or mobile phone
- Internet connection required

2.6 Constraints

- Web-based delivery only, no native app
- Must be completed within the college project timeline
- Three team members, each owning a separate module
- Free-tier tools only (MongoDB Atlas free tier, free hosting)

2.7 Assumptions and Dependencies

- Students have access to a device and internet connection
- Senior students are willing to volunteer time to conduct sessions
- Meeting links (Google Meet/Zoom/Teams) are created manually by the mentor and pasted into the session form
- MongoDB Atlas free tier is sufficient for the scale of a college project

3. Specific Requirements

3.1 Module 1: Authentication & Profile Management

3.1.1 Description

Handles account creation, login, role-based access, and profile management. A student's role (junior or senior) is decided automatically from their year of study, not chosen manually, so it cannot be misused.

3.1.2 Functional Requirements

ID	Requirement	Priority
AUTH-F1	The system shall allow a student to register with name, email, password, college, and year.	High
AUTH-F2	The system shall hash the password using bcrypt before storing it.	High
AUTH-F3	The system shall set the role automatically: year 1-2 = junior, year 3-4 = senior.	High
AUTH-F4	The system shall allow a registered user to log in with email and password.	High
AUTH-F5	The system shall issue a JWT token on successful login.	High
AUTH-F6	The system shall restrict protected pages and APIs to logged-in users only.	High
AUTH-F7	The system shall restrict role-specific actions (e.g., creating a session) to the correct role.	High
AUTH-F8	The system shall allow a user to view and edit their profile (bio, skills, social links, photo).	Medium
AUTH-F9	The system shall display the user's points and rank on their profile.	Medium

3.1.3 Non-Functional Requirements

ID	Category	Requirement
AUTH-N1	Security	Passwords shall never be stored or sent back to the browser in plain text.
AUTH-N2	Security	Every protected request shall require a valid JWT token in the Authorization header.
AUTH-N3	Usability	The login error message shall not reveal whether the email or the password was wrong.
AUTH-N4	Performance	Login and signup shall respond within 2 seconds under normal load.
AUTH-N5	Reliability	The system shall prevent two accounts from registering with the same email.

3.2 Module 2: Session Management

3.2.1 Description

Lets a senior create and manage learning sessions, and lets a junior discover, filter, and register for them. Seat limits and duplicate registration are enforced automatically.

3.2.2 Functional Requirements

ID	Requirement	Priority
SESS-F1	The system shall let a senior create a session with title, subject, category, difficulty, date, time, meeting link, and maximum seats.	High
SESS-F2	The system shall let a senior edit or delete only their own sessions.	High
SESS-F3	The system shall display all upcoming sessions to students.	High
SESS-F4	The system shall let a student search and filter sessions by subject, category, and difficulty.	High
SESS-F5	The system shall let a junior register for a session.	High
SESS-F6	The system shall prevent a student from registering twice for the same session.	High
SESS-F7	The system shall prevent registration once the seat limit is reached.	High
SESS-F8	The system shall prevent a mentor from registering for their own session.	Medium
SESS-F9	The system shall let a student view their registered and created sessions under “My Sessions”.	High
SESS-F10	The system shall let a senior view the list of students registered for their session.	Medium

3.2.3 Non-Functional Requirements

ID	Category	Requirement
SESS-N1	Reliability	The seat count shall always match the number of registered students.
SESS-N2	Performance	The session list shall load within 2 seconds for up to 100 sessions.
SESS-N3	Security	Only the session's own mentor can edit or delete it.
SESS-N4	Usability	Filter results shall be clearly labelled, with an empty-state message when nothing matches.

3.3 Module 3: Quiz

3.3.1 Description

Lets a senior create a quiz linked to a session, and lets a junior attempt it after registering. Scoring is automatic and happens on the server, so answers cannot be seen in advance.

3.3.2 Functional Requirements

ID	Requirement	Priority
QUIZ-F1	The system shall let a senior create a quiz linked to a session, with multiple questions.	High
QUIZ-F2	Each question shall have multiple options, one correct answer, and a mark value.	High

ID	Requirement	Priority
QUIZ-F3	The system shall let a junior attempt a quiz only for a session they registered for.	High
QUIZ-F4	The system shall not reveal correct answers before the student submits the quiz.	High
QUIZ-F5	The system shall calculate the score automatically after submission.	High
QUIZ-F6	The system shall show the student their score, the correct answers, and explanations after submission.	Medium
QUIZ-F7	The system shall award points automatically based on quiz performance.	High
QUIZ-F8	The quiz attempt shall be timed, with auto-submit when time runs out.	Medium

3.3.3 Non-Functional Requirements

ID	Category	Requirement
QUIZ-N1	Security	Correct answers shall never be sent to the browser before submission.
QUIZ-N2	Accuracy	Score calculation shall happen on the server, not in the browser.
QUIZ-N3	Reliability	The timer shall continue correctly even if the student refreshes the page.
QUIZ-N4	Performance	The score shall be shown immediately after submission.

3.4 Module 4: Leaderboard & Gamification

3.4.1 Description

Tracks points earned by students and mentors, and displays ranked leaderboards. Points are awarded automatically by other modules (e.g., Quiz) and totalled here.

3.4.2 Functional Requirements

ID	Requirement	Priority
LEAD-F1	The system shall award points for actions such as completing a quiz.	High
LEAD-F2	The system shall maintain a running point total for each student.	High
LEAD-F3	The system shall display a ranked leaderboard of students by points.	High
LEAD-F4	The system shall display a separate leaderboard of mentors ranked by average rating.	Medium
LEAD-F5	The system shall assign a badge to a student based on their point range.	Low
LEAD-F6	The system shall highlight the logged-in user's own position on the leaderboard.	Medium

3.4.3 Non-Functional Requirements

ID	Category	Requirement
LEAD-N1	Accuracy	The leaderboard shall always reflect the latest point totals.
LEAD-N2	Performance	The leaderboard shall load within 2 seconds for up to 500 students.
LEAD-N3	Fairness	Point values for each action shall be fixed and documented, not random.

3.5 Module 5: Ratings & Feedback

3.5.1 Description

Lets a junior rate a mentor after attending their session. Prevents duplicate or self-ratings, and keeps the mentor's average rating up to date automatically.

3.5.2 Functional Requirements

ID	Requirement	Priority
RATE-F1	The system shall let a junior rate a mentor after attending a registered session.	High
RATE-F2	The rating shall cover subject knowledge, explanation quality, communication, usefulness, and an overall score (1 to 5 each).	High
RATE-F3	The system shall prevent a student from rating the same session more than once.	High
RATE-F4	The system shall prevent a mentor from rating their own session.	High
RATE-F5	The system shall recalculate the mentor's average rating after every new rating.	High
RATE-F6	The system shall let a student leave optional written feedback.	Medium
RATE-F7	A mentor's ratings and feedback shall be viewable by other students.	Medium

3.5.3 Non-Functional Requirements

ID	Category	Requirement
RATE-N1	Reliability	One rating per student per session shall be enforced at the database level.
RATE-N2	Usability	The rating form shall be simple, using a 1-to-5 scale for each category.
RATE-N3	Security	Only students who registered for the session can rate it.

3.6 Module 6: Resource Sharing

3.6.1 Description

Lets students and mentors share links to study material — notes, PDFs, videos, or code repositories — in a searchable, central library.

3.6.2 Functional Requirements

ID	Requirement	Priority
RES-F1	The system shall let a user share a resource link with a title, description, subject, and type (PDF, notes, video, GitHub, etc.).	High
RES-F2	The system shall let a student search and filter resources by subject or type.	High
RES-F3	The system shall let a student upvote a useful resource.	Medium
RES-F4	The system shall let a student bookmark a resource for later.	Low
RES-F5	The system shall display who uploaded a resource and when.	Medium

3.6.3 Non-Functional Requirements

ID	Category	Requirement
RES-N1	Usability	The resource list shall be easy to scan and search.
RES-N2	Reliability	An invalid or broken link shall not crash the resource page.
RES-N3	Security	Only logged-in users can upload or upvote a resource.

4. System-Wide Requirements

4.1 User Interface

- Simple, clean, and modern screens
- Separate dashboards for Junior and Senior roles
- Responsive layout for laptop, tablet, and mobile
- Visible navigation on every page (menu bar with links to all major sections)

4.2 Common Requirements

ID	Category	Requirement
SYS-1	Authentication	Login with role-based access: junior, senior.
SYS-2	Privacy	Only the details required for the platform shall be collected from students.
SYS-3	Security	Passwords stored using bcrypt hashing; sessions managed using JWT.
SYS-4	Availability	The application shall be accessible through a web browser whenever the server is running.
SYS-5	Portability	The application shall work on the latest versions of Chrome, Edge, and Firefox.
SYS-6	Responsiveness	The interface shall adapt correctly to laptop and mobile screen sizes.
SYS-7	Modularity	Each of the six modules shall be built and tested independently before integration.
SYS-8	Data Backup	The database shall be hosted on MongoDB Atlas, which provides automatic cloud backup.

5. Technical Requirements

5.1 Technology Stack

Layer	Technology used
Frontend	React with Vite, React Router, Axios
Backend	Node.js with Express.js
Database	MongoDB with Mongoose
Authentication	JWT (JSON Web Token) with bcrypt password hashing
Development Tools	VS Code, Git, GitHub, Postman, MongoDB Compass
Hosting (planned)	Backend on Render, Frontend on Vercel, Database on MongoDB Atlas

5.2 Main Data Entities

Entity	Used in	Key data
User	Module 1 (all modules reference it)	Name, email, password, role, college, year, bio, skills, points, averageRating
Session	Module 2	Title, subject, category, difficulty, date, time, meeting link, mentorId, registeredStudents, max seats
Quiz, QuizResult	Module 3	Questions, options, correct answers, marks, student's score, attempt time
Rating	Module 5	sessionId, mentorId, studentId, five rating scores, feedback text
Resource	Module 6	Title, description, type, subject, link, uploadedBy, upvotes

5.3 Hardware and Software

Item	Requirement
Client	Laptop, tablet, or phone with a modern browser
Network	Basic broadband or mobile data
Server	MongoDB Atlas (database) plus Render/Vercel (application hosting)

Appendix A: Open Issues

ID	Question	Status
OI-1	Is an Admin Panel included in Version 1, or future scope?	Open
OI-2	Is the AI doubt-clearing chatbot included in Version 1, or future scope?	Open
OI-3	Is an in-app Notifications system included in Version 1, or future scope?	Open
OI-4	Will the project be deployed online (Render/Vercel), or demoed on localhost only?	Open
OI-5	College name, department, and individual team member names to be filled in the Document Control table.	Open
OI-6	Exact point values for each gamified action (quiz completion, session attendance, etc.) to be finalised and documented here once agreed.	Open